

Cambridge (CIE) IGCSE Additional Maths



Your notes

Coordinate Geometry of the Circle

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Your notes

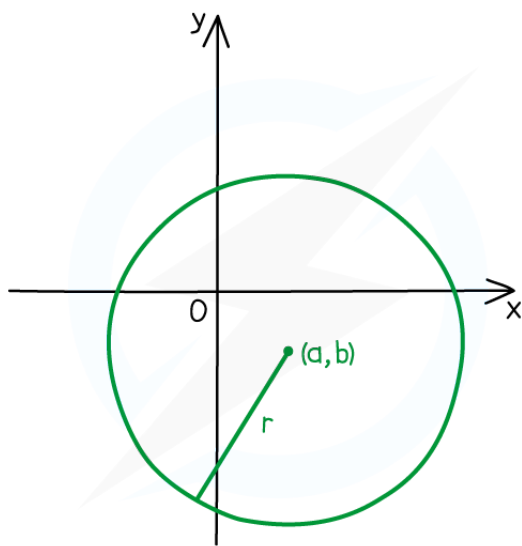
Equation of a Circle

Equation of a Circle

What is the equation of a circle?

- A circle with centre (a, b) and radius r has the equation

$$(x - a)^2 + (y - b)^2 = r^2$$



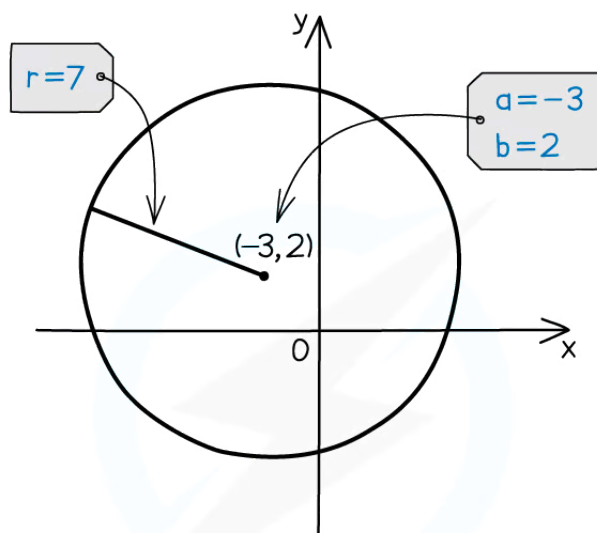
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- You need to be able to find the equation of a circle given its centre and radius
 - Substitute the values into the formula



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THE EQUATION OF THE CIRCLE IS

$$(x - (-3))^2 + (y - 2)^2 = 7^2$$

$$(x + 3)^2 + (y - 2)^2 = 49$$

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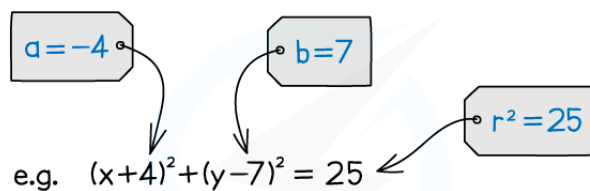


How do I find the centre and radius of a circle given its equation?

- Make sure it is in the form $(x - a)^2 + (y - b)^2 = r^2$
 - The **radius** is the **positive square root** of the constant term
 - The **coordinates of the centre** can be found by finding the values that make **each bracket equal to zero**



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e.g. $(x+4)^2 + (y-7)^2 = 25$

THE CENTRE IS $(-4, 7)$

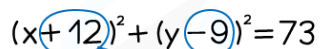
THE RADIUS IS $r = \sqrt{25} = 5$

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Examiner Tips and Tricks

- Remember that the numbers in the brackets have the opposite signs to the coordinates of the centre



$(x+12)^2 + (y-9)^2 = 73$

CENTRE $(-12, 9)$

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- Don't forget to take the **square root** of the right-hand side of the equation when finding the radius



Worked Example



Your notes



A circle has centre $(-5, -7)$ and goes through the point $(1, 1)$.
Find the equation of the circle.

THE RADIUS IS THE DISTANCE FROM
 $(-5, -7)$ TO $(1, 1)$

$$r = \sqrt{(1 - (-5))^2 + (1 - (-7))^2}$$

$$= \sqrt{6^2 + 8^2} = \sqrt{100} = 10$$

THE DISTANCE BETWEEN
TWO POINTS

(x_1, y_1) AND (x_2, y_2) IS

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

THE EQUATION OF THE CIRCLE IS

$$(x - (-5))^2 + (y - (-7))^2 = 10^2$$

$$(x + 5)^2 + (y + 7)^2 = 100$$

$$a = -5$$

$$b = -7$$

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Finding the Centre & Radius

What are the different forms of the equation of a circle?

- The most useful equation of a circle is $(x - a)^2 + (y - b)^2 = r^2$
 - This is so the centre, (a, b) and radius r are easy to see
- Any other form of the equation of a circle can be rearranged into this form
 - The most common alternative form for the equation of a circle is called the general form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

How do I find the centre and radius of a circle from any form of its equation?

- A circle equation in a different form can always be rearranged into $(x - a)^2 + (y - b)^2 = r^2$
 - The centre is then (a, b) and radius r

- Rearranging to this form will often involve **completing the square**

e.g. $x^2 - 6x + y^2 + y - 15 = 0$

$$x^2 - 6x = (x-3)^2 - 9$$

$$y^2 + y = (y + \frac{1}{2})^2 - \frac{1}{4}$$

COMPLETING
THE SQUARE

SO

$$(x-3)^2 - 9 + (y + \frac{1}{2})^2 - \frac{1}{4} - 15 = 0$$

$$(x-3)^2 + (y + \frac{1}{2})^2 - \frac{97}{4} = 0$$

$$(x-3)^2 + (y + \frac{1}{2})^2 = \frac{97}{4}$$

RADIUS $\sqrt{\frac{97}{4}} = \frac{\sqrt{97}}{2}$

CENTRE $(3, -\frac{1}{2})$

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Your notes



Worked Example



Your notes



A circle has the equation

$$x^2 + 2x + y^2 - 6y + k = 0$$

and passes through the point $(5, 12)$.

Find the centre and radius of the circle.

CIRCLE PASSES THROUGH $(5, 12)$ SO

$$(5)^2 + 2(5) + (12)^2 - 6(12) + k = 0$$

$$25 + 10 + 144 - 72 + k = 0$$

$$k = -107$$

USE THE COORDINATES OF THE KNOWN POINT TO FIND THE VALUE OF k

EQUATION OF THE CIRCLE IS

$$x^2 + 2x + y^2 - 6y - 107 = 0$$

$$(x+1)^2 - 1 + (y-3)^2 - 9 - 107 = 0$$

$$(x+1)^2 + (y-3)^2 = 117$$

REARRANGE INTO $(x-a)^2 + (y-b)^2 = r^2$ FORM

CENTRE IS $(-1, 3)$

RADIUS IS $\sqrt{117}$

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Intersection of a Circle & a Line

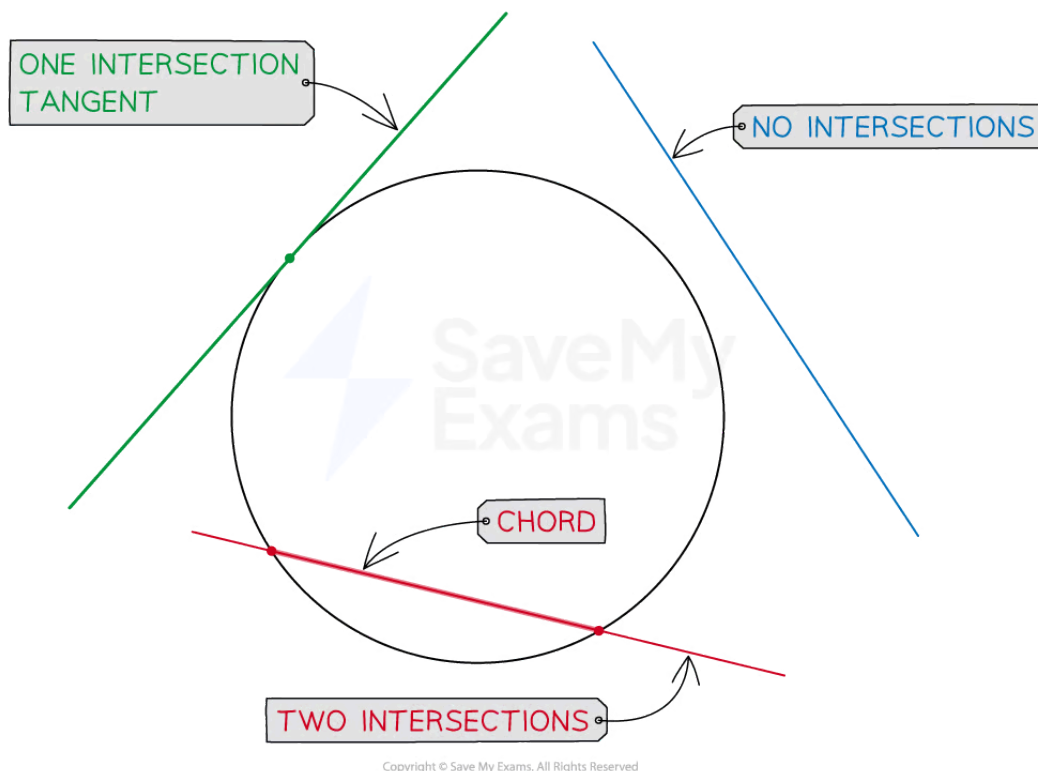
What is meant by the intersection of a circle and a line?

- A **line** may pass **through** a circle
 - in which case it will **intersect** the circle **twice**
 - the part of the line between the two points of intersection will be a **chord**
 - or, if it passes through the **centre** of the circle, a **diameter**
- A **line** may **touch** a circle
 - in which case it will **intersect** the circle **once**

- such a line would be called a **tangent** to the circle
- A line may **not intersect** a circle at all



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How do I determine whether a line and a circle intersect?

- For the equation of a circle in the form $(x - a)^2 + (y - b)^2 = r^2$ and the equation of a line in the form $y = mx + c$
- **STEP 1**
Substitute the linear equation into the circle equation
 - e.g. $(x - 5)^2 + (y - 2)^2 = 13$ and $y = x - 4$ would become $(x - 5)^2 + ((x - 4) - 2)^2 = 13$
- **STEP 2**
Expand, rearrange and simplify this equation - it should be a quadratic
 - e.g. $(x - 5)^2 + (x - 6)^2 = 13$ $x^2 - 10x + 25 + x^2 - 12x + 36 - 13 = 0$
 $x^2 - 11x + 24 = 0$



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STEP 3

Solve the equation to deduce the number of intersections

If there are **two solutions**, there are **two intersections**, **one solution** (repeated) indicates a **tangent**, **no (real) solutions** indicates **no intersection**

▪ e.g. $(x-3)(x-8)=0$ Two solutions so the line and the circle intersect twice
 $x=3, x=8$

STEP 4 If required, find the y -coordinates of the intersection(s)

▪ e.g.

$$y = x - 4$$

$$y = 3 - 4 = -1, y = 8 - 4 = 4$$

The line and the circle intersect at the points $(3, -1)$ and $(8, 4)$



Examiner Tips and Tricks

- A horizontal or vertical line could intersect a circle, when the full method shown above is unnecessary
 - Horizontal lines have the form $y = k$, so substitute k for y in the circle equation (and solve for x)
 - Vertical lines have the form $x = k$, so substitute k for x in the circle equation (and solve for y)



Worked Example

Show that the line $y = 4x + 4$ is tangent to the circle $(x-4)^2 + (y-3)^2 = 17$. State the coordinates of the point of intersection between the tangent and the circle.

STEP 1 - Substitute the linear equation into the circle equation

$$(x-4)^2 + (4x+4-3)^2 = 17$$

STEP 2 - Rearrange to a quadratic



Your notes

$$\begin{aligned}(x-4)^2 + (4x+1)^2 - 17 &= 0 \\ x^2 - 8x + 16 + 16x^2 + 8x + 1 - 17 &= 0 \\ 17x^2 &= 0\end{aligned}$$

STEP 3 - Solve

$$x^2 = 0$$

$$x = 0 \text{ (repeated)}$$

There is only one point of intersection so the line $y = 4x + 4$ is tangent to the circle

$$(x-4)^2 + (y-3)^2 = 17$$

STEP 4 - The coordinates are required

$$y = 4(0) + 4 = 4$$

The line and circle intersect at $(0, 4)$



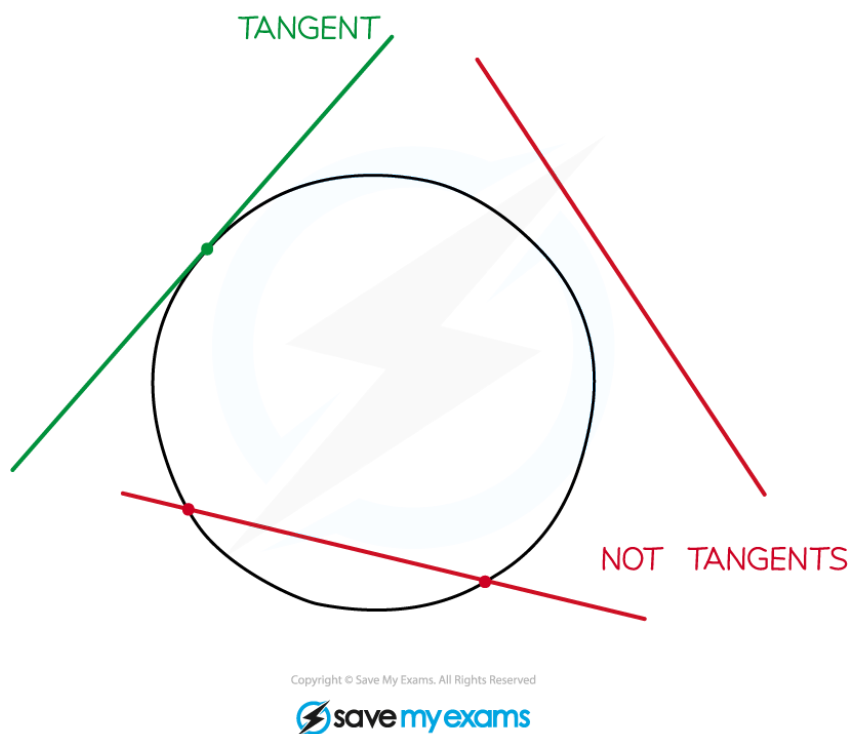
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Tangents to Circles

Tangents to Circles

What is a tangent to a circle?

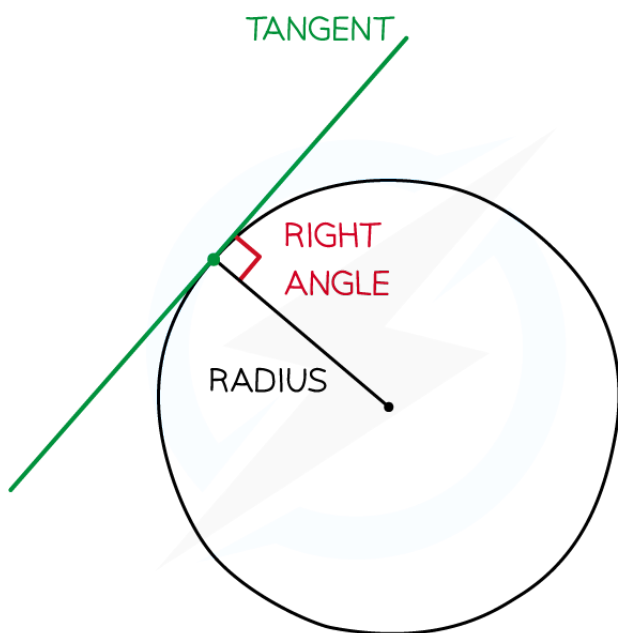
- A **tangent** (to a circle) is a line that **touches** the circle at a **single point** but does *not* cut across the circle



- A **tangent** to a circle is **perpendicular** to the **radius** of the circle at the point of intersection



Your notes



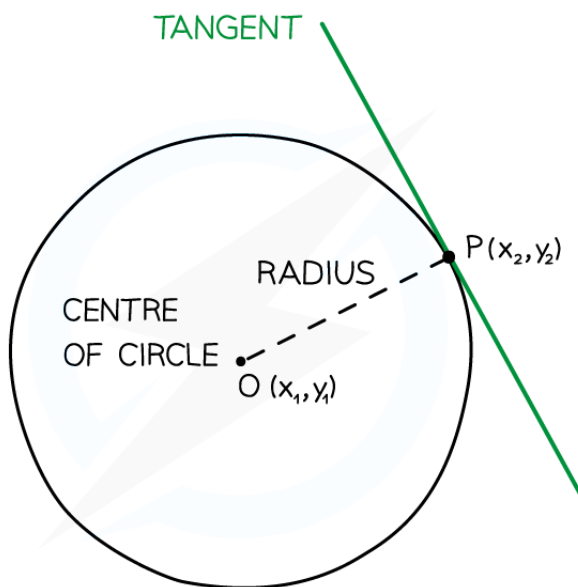
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How can I find the equation of the tangent line to a circle at a given point?



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- **STEP 1** Find the **gradient** of the **radius OP**

$$\text{GRADIENT OF RADIUS} = \frac{y_2 - y_1}{x_2 - x_1}$$

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- **STEP 2** Find the **gradient** of the **tangent** As they are perpendicular, if the gradient of the radius is m_r ,

then, the gradient of the tangent is $m_t = -\frac{1}{m_r}$ (Negative reciprocal) Alternatively,



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THE TANGENT IS PERPENDICULAR TO THE RADIUS, SO:

$$\text{GRADIENT OF TANGENT} = -\frac{x_2 - x_1}{y_2 - y_1}$$

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- **STEP 3** The **equation** of the **tangent** is the equation of the line with that **gradient** that goes through point **P** (x_2, y_2) Write down the equation using $y - y_2 = m_t(x - x_2)$ where m_t is the gradient



Examiner Tips and Tricks

- If you understand the formula in Step 2 above

$$\text{Gradient of tangent} = -\frac{x_2 - x_1}{y_2 - y_1}$$

you can find the gradient of the tangent directly without having to find the gradient of the radius first.



Worked Example



Your notes



The circle C has equation $(x - 5)^2 + (y - 2)^2 = 25$.

The point $P(2, 6)$ lies on C .

Find an equation for the tangent to C at point P .

STEP 1: C HAS CENTRE $(5, 2)$

GRADIENT OF RADIUS AT $P = \frac{6-2}{2-5} = -\frac{4}{3}$

$$\frac{y_2 - y_1}{x_2 - x_1}$$

STEP 2: GRADIENT OF TANGENT AT $P = \frac{3}{4}$

$$-\frac{x_2 - x_1}{y_2 - y_1}$$

STEP 3: $y = \frac{3}{4}(x - 2) + 6$
 $= \frac{3}{4}x - \frac{3}{2} + 6$

LINE THROUGH $(2, 6)$
 WITH GRADIENT $\frac{3}{4}$

THE EQUATION OF THE TANGENT IS

$$y = \frac{3}{4}x + \frac{9}{2}$$

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Your notes

Intersection of Two Circles

Intersection of Two Circles

What is meant by the intersection of two circles?

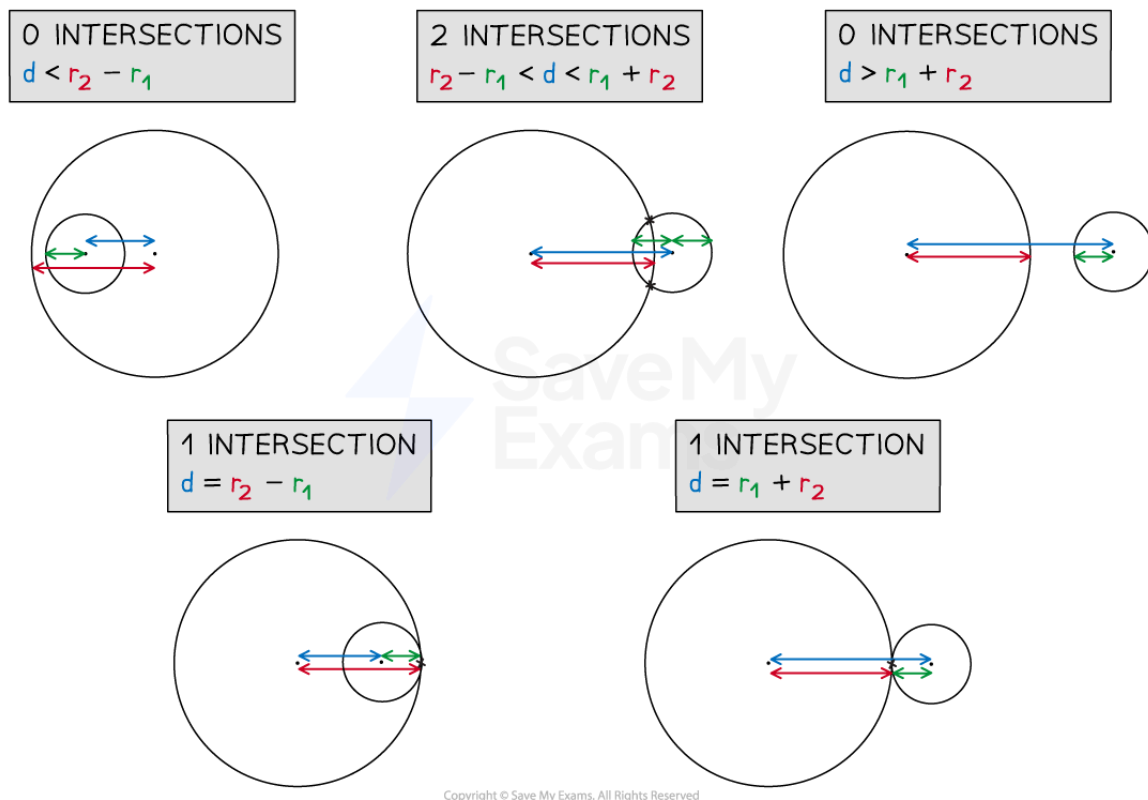
- Two circles may **intersect** once (touch), twice (cross), or not at all
 - Touching circles may be referred to as **tangent** to each other
 - they would have a **common tangent line**

How do I determine if two circles intersect or not?

- Find the distance, d , between the centres of the two circles
 - This can be found using Pythagoras' theorem
 - For centres (x_1, y_1) and (x_2, y_2) , $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$
- The radii of the two circles, r_1 and r_2 , where $r_2 \geq r_1$ are also needed
- If $r_2 - r_1 < d < r_1 + r_2$ then the circles intersect **twice**
- If $d = r_2 - r_1$ or $d = r_1 + r_2$ then the circles intersect **once**
- If $d > r_1 + r_2$ or $d < r_2 - r_1$ then the circles do **not** intersect



Your notes



- Rather than trying to remember those formulae, try to understand the logic behind each situation

How do I find the coordinates of the point(s) of intersection of two circles?

- Once it has been determined that the circles do intersect at least once, the following process can be used to determine the coordinates of any intersections
- **STEP 1 Rearrange both** circle equations so that one side is **zero**
- **STEP 2** Put the circle equations equal to each other (i.e. solve simultaneously!)
- **STEP 3** Expand/rearrange/simplify into a **linear** equation
 - The x^2 and y^2 terms will **cancel**, leaving an equation of the form $y = mx + c$, $x = k$ or $y = k$ (These are 'diagonal line', 'vertical line' and 'horizontal line') The intersection(s) will lie on this line
- **STEP 4 Substitute** the **linear** equation into **either** of the **circle** equations Solving this equation will lead to either the **X**-coordinate(s) or **Y**-coordinate(s) of the intersection(s)



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- **STEP 5 Substitute** the X (or Y) coordinates into either circle equation to find the corresponding Y (or X) coordinates. This step will not be needed in the case of the linear equation being of the form $x = k$ or $y = k$.



Examiner Tips and Tricks

- Even if not given, or asked for, a sketch of the circles can help visualise their positions relative to each other.
- You can then see if your final answers make sense with your sketch.



Worked Example

a) Determine the number of intersections between the circles with equations $x^2 + y^2 = 2$ and $(x - 4)^2 + y^2 = 10$.

$x^2 + y^2 = 2$ has centre $(0, 0)$ and radius $\sqrt{2}$.

$(x - 4)^2 + y^2 = 10$ has centre $(4, 0)$ and radius $\sqrt{10}$.

$$d^2 = (4 - 0)^2 + (0 - 0)^2$$

$$d = 4$$

Using a sketch may help you to 'see' that $d = 4$.

$$r_1 = \sqrt{2}, r_2 = \sqrt{10}$$

Compare d with the sum and difference of r_1 and r_2 .

$$r_1 + r_2 = 4.576 \dots$$

$$r_2 - r_1 = 1.748 \dots$$

$$\therefore r_2 - r_1 < d < r_1 + r_2$$

The circles intersect twice



Your notes

b) Determine the coordinates of any intersections between the circles with equations $x^2 + y^2 = 2$ and $(x - 4)^2 + y^2 = 10$.

STEP 1 - Rearrange both equations so zero is on one side

$$\begin{aligned}x^2 + y^2 - 2 &= 0 \\(x - 4)^2 + y^2 - 10 &= 0\end{aligned}$$

STEP 2 - Put the equations equal to each other

$$x^2 + y^2 - 2 = (x - 4)^2 + y^2 - 10$$

STEP 3 - Expand and rearrange until in linear form

$$\begin{aligned}x^2 + 8 &= x^2 - 8x + 16 \\8x - 8 &= 0 \\x &= 1\end{aligned}$$

STEP 4 - Substitute into either circle equation

$$\begin{aligned}(1)^2 + y^2 &= 2 \\y^2 &= 1 \\y &= 1, y = -1\end{aligned}$$

STEP 5 - Not required in this case

The intersections of the two circles have coordinates (1, 1) and (1, -1)

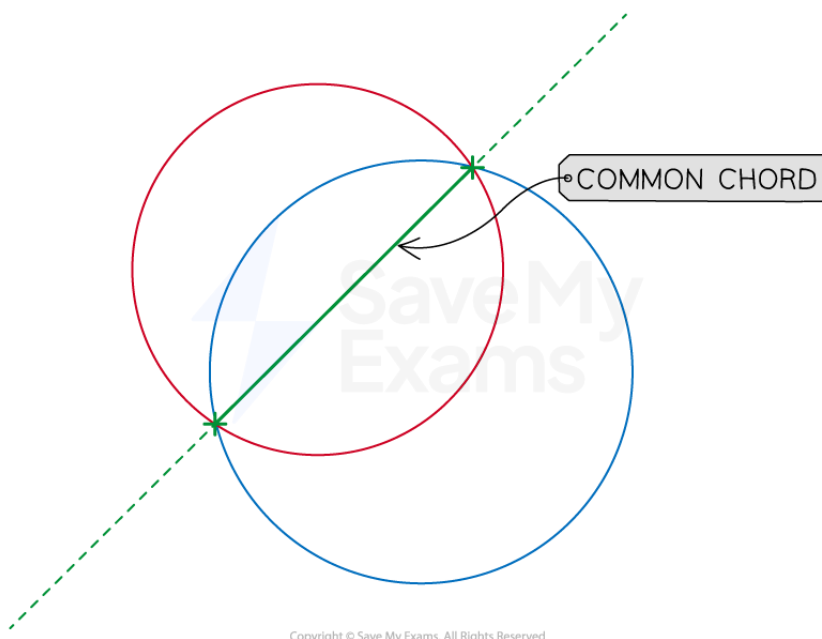
Equation of Common Chord

What is a common chord?

- For circles that **intersect twice** the **common chord** is the line that joins the points of intersection
- This line is a **chord** in both circles
 - Circles that intersect once (touch) have a common tangent



Your notes



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How do I find the equation of a common chord?

- As a **common chord** is a **straight line**, its equation will be of the form $y = mx + c$ unless
 - it is a **horizontal** line, in which case its equation will be of the form $y = k$
 - it is a **vertical** line, in which case its equation will be of the form $x = k$
- Depending on the known information, there are two ways to find the **equation** of the **common chord**
 - If the **equations** of the **circles** are known
 - **Equate** the equations and rearrange the equation into one of the three forms above
 - For example, $x^2 + y^2 - 4x + 2y - 8 = x^2 + y^2 - 8x - 2y + 16$ $4x + 4y = 24$ So the equation of the common chord is $y = 6 - x$
 - If the points of **intersection** are known
 - Use the method of finding the **equation** of a **straight line** from **two known points**
 - If the intersection points are (x_1, y_1) and (x_2, y_2) then the equation of the common chord would be



Your notes

- $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

- $\frac{y_2 - y_1}{x_2 - x_1}$ is the gradient and it can be easier to work this out first, separately



Worked Example

Two circles intersect at the points with coordinates $(3, -1)$ and $(8, 4)$.

Find the equation of the common chord of the two circles.

The points of intersection are known. Use the method of finding the equation of a straight line from two known points.

First find the gradient,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (-1)}{8 - 3} = \frac{5}{5} = 1$$

Apply $y - y_1 = m(x - x_1)$ to get the equation of the common chord.

$$\begin{aligned}y - (-1) &= 1(x - 3) \\y + 1 &= x - 3\end{aligned}$$

The equation of the common chord is **$y = x - 4$**